07-09-2026

WINDOWS SERVER

* Deploying and managing windows server 2019.
* Introduction to active directory domain services.
* Implementing IPV4.
* Managing user and service accounts.
* Implementing dynamic host configuration protocol.
* Deploying and maintaining server images.
* Monitoring Windows server 2019.

OPERATING SYSTEM - OS (SYSTEM SOFTWARE)

CLIENT OS
(End users)

- WINDOWS 10/11

SERVER OS
(physical device)
↳ server production

VERSIONS
- WINDOWS SERVER 2019
- WINDOWS SERVER 2022
- WINDOWS SERVER 2025

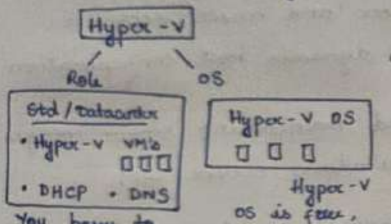
SERVER:
- HARDWARE SERVER
- SERVER OS

SERVICES: HOSTING
- DNS SERVICE
- DHCP SERVICE
- ADDS
- WEB SERVER
- MAIL SERVER
(provided by server OS)

CLIENT OS: it only request for services.
SERVER OS: it is capable of requesting & offering.

WINDOWS SERVER 2019 EDITIONS (4 Editions)

- → Windows server 2019 $\boxed{\text{Essential}}$ - 25 users, upto 50 devices can connect (no VM available) → basics virtualization or containerisation → not available
- → Windows server 2019 standard } using Hyper-V role we can create VM's
- → Windows server 2019 datacenter }
- → Hyper - V server 2019 - OS + virtualization combined, free Hyper - V OS, provides CLI



You have to pay licensing cost for standard / datacentres as well as OS inside VM's

Hyper-V OS is free, just the cost for OS in VM's.

Standard

(Buy 1 std license, get 2 VM + OS free)
STD windows server
→ pay extra and can use other VM.

Datacenter

(Unlimited VM's on the same box for free)
Buy 1 license get unlimited VM's
→ Expensive license.
→ 10% extra features compared to std.
→ More powerful hardware & core server

- → Lakhs of users are supported in standard & datacenter.
- → Same editions are available in different version eg: Windows server 2019/2022/2025 across all.
- → ~~In~~ ~~w~~ From 2022 SERVER, Windows server 2022 datacentre : Azure Edition got added (total 5 edition) and the same continues to 2025 Edition.

Benefits of Windows server azure edition.

- → Hotpatching - no restart req. during VM update.
- → SMB over QUIC - file share over internet securely without traditional VPN
- → Storage replica compression - data packets are compress during replication.
- → Extended network for Azure - On-premises network segments stretched onto Azure (cloud) for hybrid integration.

Hardware Requirements for Windows server 2019.

Component	Requirement
Processor Architecture	64 bit
Processor speed	1.4 gigahertz (GHz)
RAM	512 MB (2GB - recommended)
Hard drive space	32GB

- → All the server OS from 2016 are of 64 bit.

LAB NOTE: → .iso extention files are directly readable by VM's (extract, create new media, copy).

- → Host - only networking - for full isolation.
  ↳ (Network type) - own private network.
- → $\boxed{\text{Ctrl- G}}$ + Enter to get cursor onto VM.
- → While installing OS if you are not able to boot check the ISO image file in VM settings.
- → Installation options / methods

※ { 1> Full server (Desktop experience) GUI } Std/DC.
  2> Server core (CLI)

(AM)
Remote Management -

→ With WinRM (by default), you can use consoles, command-line utilities, or Windows PowerShell to perform remote management tasks.
→ RDP - Remote Desktop Protocol (entire machine) ↳ additional license.
→ Remote Assistance - QuickAssist (tool) [Help desk]
→ RSAT - Remote Server Administration tool : used to connect to server and manage.
  ↳ Application installed on client machine, you can manage the server without login to server. (some parts)

LAB STEPS:

→ Windows icon left click → find server manager
→ Server Manager → Manage → Add roles & features
                  ↳ Remove roles & features
→ Server Manager → Local server → Events.
                              → Services
→ Bootloop - to avoid right click on DVD drive (This PC) and click Eject
→ Slmgr — runas administrator.
  • "Slmgr.vbs -rearm" in elevated command prompt
    ↳ restart computer OS - to reset the expired license (2 times) trial period.
  • Slmgr /dli - license info
  • Slmgr /xpr - expiry date

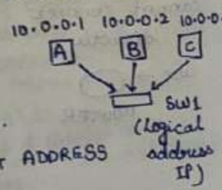
→ If Home is not there, Goto file → New VM (in VMWorkstation)
→ No media is displayed - .iso image is not mounted.
→ Goto client → right click → settings → CD/DVD → Use ISO image file. (make sure connected option is checked) if not loading.
→ Server / Client OS - system reserved partition (upto 500 MB)
  (only for 1st os) → default partition
  ↳ some of OS resource
  ↳ Bitlocker info
→ Client OS - Windows 10/11 Editions
  • Home (home users)
  • Pro • Enterprise — Domain Join feature exclusive.

IP - INTERNET PROTOCOL
  ↳ IPV6 128 bit HEXA DECIMAL
  ↳ IPV4 32 bit BINARY

IPV4 - Dotted decimal notation

0 ✗ 1-126 128-191 192-223	0-255	0-255	0X 0-255
8bits	8bits	8bits	8bits

00000000 = 0
11111111 = 255] $2^8 = 256$ (including 0)



CLASS :
A: 1-126
B: 128-191
C: 192-223 } Directly Assign.
D: 224-239 MULTICAST ADDRESS
E: 240-254 R & D

SPECIAL NUMBERS:

0 : 192.168.10.0 → SUBNET ID / NETWORK ID → 0-255
Ends with 0.
192.168.10.5

127 : LOOPBACK ADDRESS FOR SELF TEST (CAN'T ASSIGN) 127.0.0.0 - 127.255.255.255

Server Core (CLI)
→ less resource intensive OS
→ more secured
→ less updates
※ → locally managed using "Sconfig"
→ remotely we manage
→ few roles limitation

※ * We cannot convert desktop experience to server core or server core to desktop experience, should be selected during installation & final.

3> Nano server : containers.

→ [Ctrl Alt] - to get cursor out of VM

LICENSING TYPES:

1> OEM - Original Equipment Manufacturer
• Non transferable to other device.

2> Retail -
• License can be tranfered from one device to another.

(only user)

3> VLK - [Volume License key]
• Provided only for companies / organisation.
• Has Master keys
• MAK - Multiple Activation keys (used in phones)
• KMS - Key management service
↳ Activation services
• Can manage multiple devices or keys.

※ Licensing and activation models for Windows server

Licensing for Windows server standard and datacenter is based on the number of cores, not processors.

• Each windows server has the following minimum license requirement :
* All physical cores must be licensed
* There must be 8 cores licenses per processor.
* There must be 16 core licenses per server.

• CAL - Client access licenses are required for each user or device that connects to the server for any purpose.

CAL
Per user / Per device

Eg: Trainer - Company - Paid | OS license.
Trainee - ask question - per person some cost - CAL

Server - OS : all comes as a package.
→ Roles - DNS, DHCP, ADDS, web server
→ Features - windows backup, .Net, group policy etc
* Roles can have features as dependancy.

Server Manager
→ We will be able to install roles & features, remove roles & features on the local server and other servers in the network.

255 : BROADCAST ADDRESS

CAASS A : 10.255.255.255
B : 162.5.255.255
C : 194.10.16.255

we can control the underlined part.

CLASS	SUBNET MASK - to know N/W bits & IP bits
A: 1-126	255.0.0.0 = 2^24 = 16777216 - 2 = 16777214
B: 128-191	255.255.0.0 = 2^16 = 65536 - 2 = 65534
C: 192-223	255.255.255.0 = 2^8 = 256 - 2 = 254

SUBNET MASK — HOST/IP bits (0's)
— NETWORK bits (1's)

COMMUNICATION RULES B/W CLASSES

RULES 1

X CLASS A — B, C

cannot connect directly

use → ROUTER

RULE 2

TO BE IN SAME SUBNET/NETWORK

- CLASS A 1st OCTET VALUE MUST BE SAME.
- CLASS B 1st & 2nd OCTET VALUE MUST BE SAME.
- CLASS C 1st, 2nd & 3rd OCTET VALUE MUST BE SAME.

A —— B

10.0.0.1 — 10.200.250.5 ✓

10.0.0.1 — 20.0.0.2 ✗ use router

→ Default Gateway - Router ID.

IP
- PRIVATE IP (LAN) — FREE | LOCAL UNIQUE
  - A: 10.0.0.0 - 10.255.255.255
  - B: 172.16.0.0 - 172.31.255.255
  - C: 192.168.0.0 - 192.168.255.255
- PUBLIC (WAN) — COST | GLOBALLY UNIQUE, ISP SUPPORT (Except private range all other address are public)

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WINDOWS NETWORK

→ SAM - Security Account Manager - to manage accounts locally.

→ WORKGROUP (DEFAULT) - decentralised approach, if one system is down, it is not accessible.

local database — TOM [PC1 W] —— [PC2 W] JACK (user account)
10.0.0.1 — 10.0.0.2

→ DOMAIN △ Eg: ST.COM (stratagent-ST) — central database

ADDS ROLE & CONFIGURE

C:\WINDOWS\NTDS - NTDS.DIT

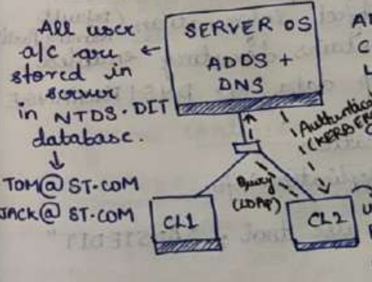
↳ OBJECT: USERS, COMPUTERS, GROUPS, ETC.

C:\WINDOWS\SYSVOL

↳ SCRIPTS & GROUP POLICY SETTINGS

All user a/c are stored in server in NTDS.DIT database.

TOM@ST.COM
JACK@ST.COM



USERNAME: TOM
PASSWORD: XXXXXX
LOGON TO: LM (Local Machine) | DOMAIN ✓

PROTOCOLS:

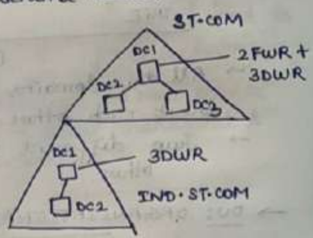
- LDAP: Querying & Communication (389)
- KERBEROS V5: Authentication (88) > port no.

* Domain user cannot logon to workgroup.

Domain Controller.

DC: Its a server that hosts NTDS. DIT & SYSVOL, it is also act as Kerberos & LDAP server.
Every domain must have 1 or more DC in domain, without DC we can't create domain.

NOTE: → Companies can create 2 seperate forest when you want seperate schema rules.



FSMO ROLES

DOMAIN WIDE ROLE	FOREST WIDE ROLE
- PDC EMULATOR	- SCHEMA
- RID MASTER	- DOMAIN NAMING
- INFRASTRUCTURE	

→ Every first DC1 has default all DWR, FWR, GC roles.

FSMO (Flexible single master operational role)

DOMAIN WIDE ROLE :

• PDC (Primary Domain Controller)
TIME SYNC | PASSWORD UPDATE | GROUP POLICY UPDATE / REPLICATION.

RID (Relative Identifier) used to assign unique ID to objects. DID + RID = SID
(Domain Identifier) (Relative identifier) (Security identifier)

INFRA: It keeps information of group membership & cross domain.

FOREST WIDE ROLE:

SCHEMA MASTER: It maintains & updates the schema changes all the domain in the forest.
DOMAIN NAMING: addition & deletion of domain in forest.

* "NETDOM QUERY FSMO" Command to view all FSMO at once.

GLOBAL CATALOG SERVER: It keep's partial info. of object through out forest in read only mode.
* GC and INFRA ROLE should not be placed on same DC.
* useful when there is more than 1 domain.

LAB SCENARIO:

* → To install active directory, DNS is a prerequisite (if you install ADDS, DNS will auto install)

① SERVER1
IP: 10·0·0·1
255·0·0·0
DNS: 10·0·0·1

Subnet ID
10·0·0·0
BroadCast ID
10·255·255·255

CLIENT1
IP: 10·0·0·2
255·0·0·0
DNS: 10·0·0·1

② Install ADDS role & Configure
Domain name: training·com

NTDS - DIT : New Technology Directory Service - Directory Information Tree.

LDAP : light weight directory access protocol

KERBEROS V5
- KDC (Key Distribution Centre)
  - TGT (Ticket Granting Ticket) → TGT is used for security, instead of typing password everytime and to prevent leaks.

Advantages (DOMAIN) : → Centralised database → More secure
→ Replication → better management
→ Access to clients

° ACTIVE DIRECTORY PARTITION ° (AD)

→ "NTDS-DIT" - partitions inside the database include
forest wide FWR
* SCHEMA - Contains object, class, attributes (properties)
* CONFIGURATION - AD topology, sites, services.
* DOMAIN PARTITION - object information (Default Naming Partition)
* APPLICATION (OPT) - contains directory enabled app data Eg: DNS | EXCHANGE
optional

Eg: SCHEMA - Blank application
DOMAIN PARTITION - Application Info.

→ To view partition we use tool : "ADSIEDIT"

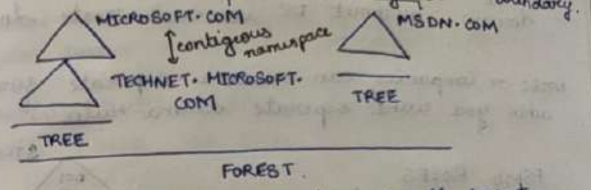
ACTIVE DIRECTORY COMPONENTS

LOGICAL
- DOMAIN
- TREE
- FOREST
- OU

PHYSICAL
- SITES
- DOMAIN CONTROLLER (DC)

LOGICAL

→ Domain - △ ← symbol - logical boundary / replication boundary in which user, computer, groups etc... are created which will also act like security & replication boundary.



→ All the domains and tree in the forest trust each other.

→ Two different forest do not trust each other.

→ OU: ORGANIZATIONAL UNIT (sub-container inside a domain)

SALES OU
- USERS
- COMP

- Deligation
- Group policy
(for giving access, permission, assign tasks)

→ Tree : contains one or more domains which forms contigeous namespace {TECHNET.MICROSOFT.COM}

→ FOREST : Contains one or more domain trees which does not form contigeous name space {MSDN.COM}

PHYSICAL

→ SITE : Contains multiple IP subnets
TWO TYPES : - INTRA SITE : Single site
- INTER SITE : Multiple site

③ Will join client 1 to domain {from workgroup to domain}

④ Create user accounts & check login
tom@training.com

Command : ncpa.cpl - to goto network adaptor. (ethernet)

Domain Join : ① Online Method -> Offline method (djoin)

② While joining to domain check : [ONLINE]
- Connectivity : both (client & server) should be connected to same n/w
- IP address - DNS IP (in the client)
(if not, could not be connected error)

NOTE : • When login to domain USERNAME: training\Harshitha

TROUBLE SHOOT
• When we join a computer to domain if we are unable to login using a account which we created on domain controller, then
→ Once login with domain name\administrator (training\administrator) it will work.
↳ and log out ⟶ then login with user

→ Domain Controller (Default OU - Organisational units)

USER TYPES:

1> LOCAL USERS
- Created on local computer
- Logon to same computer
- Password is not mandatory
- TOM (just username)

2> DOMAIN USERS
- User accounts are created on Domain Controller
- Logon to any client in domain
- Password is mandatory { 7CH | COMPLEX } (7 characters)
- TOM@TRAINING.COM > UPN (User Principle name)

* USER TYPES :
- ADMINISTRATOR : OWNER SUPER ACCOUNT
- STANDARD USER : LIMITED USER

* ADMINISTRATOR TYPES :
- LOCAL ADMIN : creating VM's, local.
- DOMAIN ADMIN : server a/c. (organisation)
- ENTERPRISE ADMIN (FOREST OWNER) : ADDS. exclusive previleges.

DHCP : DYNAMIC HOST CONFIGURATION PROTOCOL
↳ works based on lease period, and is not permanent, it changes based on conditions.

ASSIGN IP
→ STATIC / MAN (person) - usually server are considered for static IP. (permanent until we change)
→ DYNAMIC / DHCP

POOL / SCOPE - to auto. assign IP using DHCP
- STARTING IP : Eg - 10.0.0.100
- END IP : Eg - 10.0.0.200
- EXCLUDE RANGE : anything to exclude b/w start IP & end IP (numbers).
- LEASE PERIOD : 8 DAYS {DEFAULT}
OPTIONS - YES
- GATEWAY : ROUTER IP
- DNS NAME & DNS IP : training.com | 10.0.0.1
- WINS

→ DHCP will autobackup the DHCP database every 60 mins and stored in (same server)
c: /windows/system 32/DHCP ↳ Backup folder.

→ Manual backup is done, where backup info is stored outside the server for in case of physical server damage.

COMMON DHCP ISSUES

→ We can have multiple DHCP server, but it should be legitimate.

① → Address conflicts ← unique IP for all ↳ (usually in static) → ② Failure to obtain DHCP address

③ → Address obtained from incorrect scope

④ → DHCP database suffered data corruption or loss

⑤ → DHCP server has exhausted its IP address pool.

② - Condition
- Connectivity
- Scope has no IP to offer.
- Filtering (based on MAC you can filter devices not to obtain IP).

③ check the relay agent

④ use backup

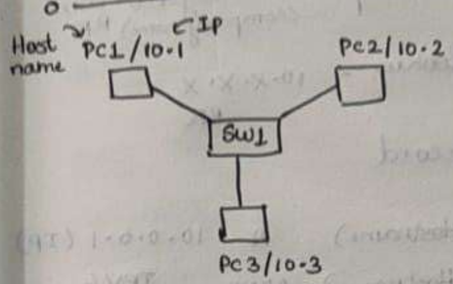
⑤ Extend the scope or create a new scope.

→ In the absence of DHCP, APIPA will be used.

→ But as soon as DHCP server is down APIPA will not be given, because only when you restart or repair your device the client recieves the APIPA.

maybe same or diff. IP
→ to refresh IP Manual lease renewal DHCP : ipconfig /release
ipconfig /renew

DOMAIN NAME SYSTEM (DNS)



PC1 : PING 10.0.0.3
(directly communicate)

PC1 : PING PC3? - now the device should search or find IP of PC3.

NAME RESOLVING SERVICES

1> DNS - Domain Name system/service: it resolves
- Hostname to IP &
- IP back to hostname

2> WINS - Windows internet naming services: it resolves NETBIOS name to IP. (legacy system).

* **Host Name** - upto 255 characters long
  - Part of FQDN (Fully Qualified domain name) CL1.training.com

* **NetBIOS Name** - max 16 characters, 15 ch represent name, 16th ch identifies service
  - flat name space (not used anymore)

PC1: PING PC3 ? to find IP of PC3, it checks.

valid for 1 hour ← 1. CLIENT CACHE — DYNAMIC / STATIC (edit via host file)
2. DNS - provides internetwork (n/w) communication.
3. NETBIOS CACHE
4. WINS
5. BROADCAST - does not work if the system is in other n/w, router does not pass broadcast messages.

DORA:

D: DISCOVER (client to server)
O: OFFER (server to client)
R: REQUEST (client to server)
A: ACK (server to client)
(Acknowledgement)

} BROADCAST

DHCP PORTS:

- 67: DHCP SERVER
- 68: DHCP CLIENT

RENEWAL:

- REQUEST & ACK (UNICAST)
1. 50% - exactly at 50% you have to renew
2. 87.5% - exactly at 87.5% you have to renew.
3. 100% - DORA will start again, if you do not renew after it reaches 100%.

→ In the absence of DHCP, clients will auto generate APIPA (client generated) (Automatic Private IP address)
↳ 169.254.X.X availability
↳ senses the (state) of DHCP server, when DHCP becomes active, original IP will be given.

RESERVATION: IP remains unchanged (but assigned through DHCP) without lease period
↳ when the DHCP server is down, APIPA (client generated) address will be assigned.
↳ after the DHCP is active, the reserved IP will be assigned back.

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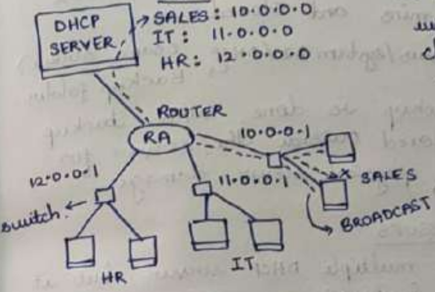
DHCP RELAY AGENT (RA) - service enabled on the router. used when more than one scope is present, ie during multiple scope is present.

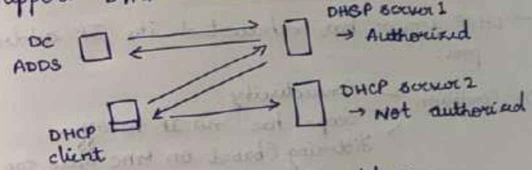
SCOPE:-
SALES: 10.0.0.0
IT: 11.0.0.0
HR: 12.0.0.0



→ In RA, DORA will repeat twice, client to agent - agent to server.

DHCP SERVER AUTHORIZATION

DHCP authorization registers the DHCP server service in the active directory domain to support DHCP clients.



DHCP client receives IP address from authorized DHCP server 1

DHCP OPTIONS: you can apply DHCP options at various levels :- Server, Scope (mostly default), class, reserved clients.

DHCP database: is a dynamic database which stores scopes, address, leases, reservation

DHCP scope Database path - C:/windows/system32/DHCP

DNS

ZONES (like database)
- Forward lookup - stratogent.com (company name) RR
- Reverse lookup - 10.x.x.x RR

RR TYPES - Resource Record

A (HOST A)	CL1 (Hostname)	A	10.0.0.1 (IP)
AAAA (HOST)	CL1 (Hostname)	AAAA	IPV6
Reverse lookup PTR (POINTER)	10.0.0.1 (IP)	PTR	CL1 (Hostname)
MX (MAIL EXCHANGE)	SVR1	MX	10.0.0.200
CNAME (ALIAS)	SVR100	CNAME	SVR1

CNAME (ALIAS)
↳ name to name mapping
↳ to add www. to website

BY DEFAULT both in F & R zone.
- NS (NAME SERVER)
  ↳ where DNS is installed
- SOA (START OF AUTHORITY)
  ↳ Keeps track of zone changes
  ↳ replication interval
  ↳ has info of host master

SRV (SERVICE LOCATION RECORD)
↳ LDAP ↳ KERBEROS ↳ GC ↳ HTTPS ↳ FTP ↳ TEAMS

DNS HIERARCY

- Root (13 root servers) ↓ A-M
  - .edu
  - .com - TLD (Top level domain)
    - stratogent.com - SLD (secondary level domain)
      - ind.stratogent.com - SD (sub domain)

www. microsoft. com .
③ ② invisible (root) ①

what is IP of microsoft?

workstation → Local DNS server → Root DNS server → .com DNS server → Microsoft.com DNS server

IP is fetched.

DNS TROUBLESHOOTING

→ nslookup
→ ipconfig /display DNS - to view client cache
→ ipconfig /flush DNS - to clear client cache
→ Always clear DNS resolver cache before troubleshooting
→ Use the hosts file for troubleshooting

NOTE: 1> DNS can be installed in workgroup (in server OS) - will be stored Database in windows / system32 / dns.

2> DNS with ADDS, database (DNS) will be stored in NTDS.DIT

→ Root hint file contain the IP addresses for DNS root servers.

DNS QUERIES

→ Queries are recursive or iterative
→ DNS clients and DNS servers initiate queries
→ DNS servers are authoritative or nonauthoritative for a namespace.
→ An authoritative DNS server for the namespace will either:
- Return the requested IP address
- Return an authoritative "no"

→ A nonauthoritative DNS server for the namespace will either:

- Check its cache
- Use forwarders
- Use root hints.

FORWARDING

A forwarder is a DNS server designated to resolve external or offsite DNS domain names.

Conditional forwarder - forwards requests using a domain name condition.

DNS Zone types

Primary - Read/write copy of a DNS database

Secondary - Read only copy of a DNS database

Active Directory Integrated - Zone data is stored in AD DS rather than in zone files.

Stub - Copy of a zone that contains only records used to locate name server.

LAB STEPS:

1→ 1> Install DHCP role & configure

Scope A : Starting IP - 10.0.0.100
End IP - 10.0.0.200
Exclude - ______

2> Enable DHCP client on client 1 & verify IP

3> Check APIPA address

4> Reservation

5> DHCP backup & restore

→ We use forwarder to forward queries to ISP's DNS or any other DNS.

→ Two scopes cannot be created with the same subnet. Eg: Scope 1 - 10.0.0.100 - 10.0.0.200
for Scope 2 - we cannot give 10.0.0.1 - 10.0.0.50
you can use 11.0.0.1

→ Address lease - computer who have taken IP from DHCP

→ Logon to Domain 2 ways:

1> training\administrator
training\username - i.e training\harshitha

2> administrator@training.com

→ Logon to local computer

.\administrator
.\harshitha.

→ Unique ID - MAC.

→ services.msc - to see & manage services.
↳ microsoft console.

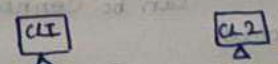
1→ DNS.

1> Explore DNS

2> Create forward lookup zone as nutraders.com

3> Configure reverse lookup zone
10.x.x.x

4> Create RR



10.0.0.50 10.0.0.51

SELECTING A FILE SYSTEM

1> FAT PROVIDES:
- → Basic file system → Partition size limitations
- → no security - hence do not use FAT

2> NTFS provides: (feature based file system)
- → Security — Permission (ACL)
  — Encryption
- → Audit → Compression
- → Disk shrink and Extend disk
  - shrink: remove extra space in disk & create new partition
  - extend: if there is any free space you can club it with other partitions extending the disk.

3> ReFS provides
- → Enhanced data verification & error correction.
- → Support for larger files, directories, volume, etc.
- → Only provides extend volume of disk and does not provide shrink disk.

→ Extend disk only happens when there is free space next to it.

DISK TYPES - HDD

- BASIC [500GB] [500GB] 500 GB two partitions
- DYNAMIC [500GB] + [1TB] → = 1.5 TB SINGLE PARTITION

DYNAMIC DISK USED RAID (Redundant array of independant / inexpensive disk)

RAID: - Performance (fast) - Redundancy (tolerate failure)

implement:
- ① Hardware (HW) TECH : RAID CONTROLLER
- ② Software TECH : SERVER OS

→ RAID combines multiple disks into a single logical unit to provide fault tolerance and performance.

→ RAID provides fault tolerance by using:
- Disk mirroring - Parity information

→ RAID can provide performance benefits by spreading disk I/O across multiple disks.

→ RAID should not replace server backups.

RAID LEVELS:

* RAID 0 : striped set without parity or mirroring (Performance based) only - auto partition handling
  - Min 2 harddrive to implement - 100% space is usable.
  - no space is dropped,

* RAID 1: Mirrored drives
  - Min and Max 2 Harddrive to implement
  - data is replicated, i.e disk 0 data is replicated onto disk 1 & hence only 50% storage used.
  - used for OS disk on servers
  - fault tolerance based.

* RAID 5: block level striped set with parity distributed across all disks.
  - min 3 hard disk - performance based
  - parity : metadata (info about actual stored data)
  - fault tolerance also provided
  - can only withstand single disk failure.

10.0.0.100 SVR1 mail server

10.0.0.101 SVR2 web server

Create → 4 Host A
→ 4 PTR
→ 1 MX
→ 2 SRV
→ 1 CNAME

5> run commands
nslookup
ipconfig / displaying & flush DNS.
ipconfig /registerdns. ipconfig /flushdns

→ Port no - Global Catalog Server 3268

→ DNS Port no - 53

→ DNS - is also a site aware application

→ Very first zone created cannot be secondary, it should be primary since in primary database is created.

→ ::1 - IPv6 loopback address.

10-09-2026

STORAGE

(HDD) Hard Drive Disk

1> Magnetic
- Performance is good
- large capacity
- low cost.
Non volatile memory

2> SSD
- best in performance
- Expensive.

3> NVMe - type of SSD where communication to motherboard happens through BUS (faster) than cable as in SSD.

express.

used in PC's: + NVMe
SATA - Serial advance technology attachment

in server:
SCSI - Small computer (parallel) system interface (8-16 devices can be connected)
SAS - Serially attached SCSI.

STEPS TO ADD NEW DISK

500GB OS+APPS+DATA

500GB

1> DISK MANAGEMENT TOOL
2> BRING DISK ONLINE
3> INITILIZE DISK
MBR (MASTER BOOT RECORD)
GPT (GUID BASED PARTITION TABLE)
4> CREATE PARTITION
5> LOAD FILE SYSTEM - FAT32 | NTFS | REFS

DISK PARTITION
- PRIMARY (Bootable support) (1st OS should be installed on primary)
- EXTENDED → LOGICAL

MBR : MAXIMUM 4 PRIMARY PARTITION
size 2TB
3 are used as primary, the 4th one can be used as extended partition with unlimited logical partition for better management. You can also use all 4 as primary.

GPT : MAXIMUM 128 PRIMARY PARTITION
size upto 18EB
1-127 (Primary) 128th - Extended with unlimited logical partition
better management

* **RAID 6:** Block level striped set with parity distributed across all disks.
  - → Dual parity → can withstand 2 disk failure at once. (7 disk req.)
* **RAID 1+0:** Each pair of disk is mirrored, then the mirrored disks are striped.
  - provides performance + fault tolerance
  - RAID 0 + RAID 1
* **Spanned volume:** combines multiple physical hard drives into a single large logical drive by writing data sequentially to one disk until it is full before moving to the next.

STORAGE SPACE: (similar to SAN)

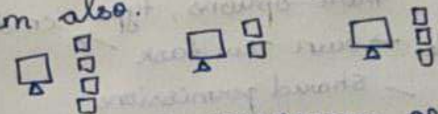
Use storage spaces to add physical disks of any type and size to a storage pool, and then create highly available virtual disks from the storage pool.

PHYSICAL DISK → STORAGE POOL → VIRTUAL DISK → DISK DRIVE

1TB, 1TB, 1TB → Storage Pool 3TB → 10GB virtual disk - specify the RAID level here → 10GB Partition - load the file system

STORAGE SPACE DIRECT:

Same concept as storage space, but the hard drive are combined from external system also.



VIRTUAL DISK CONFIGURATION OPTIONS

Feature	Options
Storage layout	- Simple - Parity - Two-way or three-way mirror (1 disk failure tolerance) (2 disk failure tolerance)
Drive Allocation	- Data store - Manual - Hot spare
Provisioning schemes	- Thin (dynamic) (VD 10GB can be extended) vs fixed provision (static) (VD - 10GB fixed)

LAB:

→ when you want to change ReFS to NTFS using format, data will be lost, you have to backup the data.

※ → $\underbrace{\text{convert E: /fs:ntfs}}_{\text{command}}$ - to change file system without data loss.

→ In storage pool, primordial will only be present if there are unused disks.

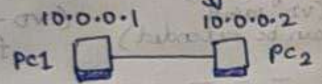
→ msinfo32 - system info on server core.

→ If command prompt is closed, when you type exit - right click on server core - send ctrl + alt + del option, select task manager, ~~more options~~, type cmd enter.
↳ File → run new task

PERMISSIONS
- Shared permissions
- NTFS permission

1> Shared permission
- folder level - volume level

sharing methods:
- SIMPLE SHARE
- ADVANCED SHARE
  - Concurrent access
  - Advanced permission
  - Offline access.



DATA1
DATA2
DATA3 $ - Hidden file

\\ 10.0.0.2 - all info except hidden file
\\10.0.0.2\DATA1 - particular file
\\10.0.0.2\DATA3$ - hidden file info.
MAP DRIVE - without using the IP commands
ADMINISTRATOR SHARED (ALL DRIVES)
\\10.0.0.2\C$
\\10.0.0.2\D$

2> NTFS permission - used for local resource
- FILE LEVEL - DRIVE / VOLUME LEVEL.
- FOLDER

Permission rules:

1> Higher permission will take precedence over lower permission.
2> Any permission with deny is directly denied.

* When we combine shared & NTFS permission together most restrictive will take precedence.

Eg: i.e if shared gave full control
NTFS gave read only
read only will be considered. (most restrictive).